2. Yellow-highlighted text on the page itself. To edit, either: • Open this file in any text editor (Notepad, VS Code) OR in the Canvas page HTML editor, use Find (Ctrl+F) for the word EDIT and replace the highlighted text; or • Paste this whole file into a Canvas Page (Edit > "" HTML view), then switch to the normal editor and type over the yellow text. The FIVE things to change are all near the top, in the info strip: (A) Facilitator name (B) Facilitator email (C) Office-hours schedule (D) Grade level (if your section differs) (E) Grading weights, IF your section's official weights differ from the defaults shown in the Assessment & Grading section. When you are done, you may DELETE this yellow "How to customize" banner and the yellow highlight color (search for background:#FFF3B0 and remove it). Nothing else needs to change from year to year except the dates.
==========
-->

⚙ Facilitator: customize before publishing (then delete this box)

Replace the yellow-highlighted text below with your own details: your **name**, **email**, **office hours**, your section's **grade level**, and — only if your section differs from the defaults — the **grading weights**. Search this file for the word **EDIT** to jump to each field. No AI needed.

Geometry Honors

OPTIMA ACADEMY ONLINE · COURSE SYLLABUS · 2026–2027

Full Year · Grades 9–12

FACILITATOR Optima Academy Online Faculty	EMAIL & OFFICE your.name@optimaed.com · 239.673.2801 (OAO Front Office)
FLORIDA COURSE Florida 1206320 · Full-year course (1.0 credit)	OFFICE HOURS Posted in Canvas · weekly study hall and VR workshops
GRADE LEVEL Grades 9–12 (typically Grade 10)	FORMAT On-Demand (also offered live on the VR campus)

About this syllabus. This is the plan for the course as designed. Geometry Honors is a full-year high-school course; On-Demand scholars pace through its eleven modules across the four quarters of their enrolled year. On-Demand and live-school sections share the same content and expectations; exact pacing, scheduled dates, and a few activities may vary from section to section and term to term. Your facilitator will confirm the specifics for your section.

Course Overview

Geometry Honors teaches scholars to reason about shape, space, and proof with the precision of a mathematician. Instruction emphasizes five areas: (1) proving and applying relationships and theorems

about two-dimensional figures using both Euclidean and coordinate geometry; (2) establishing congruence and similarity from Euclidean criteria and from rigid transformations; (3) extending geometric measurement to two- and three-dimensional figures; (4) creating and applying equations of circles in the coordinate plane; and (5) developing an understanding of right-triangle trigonometry. The work is hands-on: every lesson pairs interactive content with independent practice, many lessons add a compass-and-straightedge construction or a written proof, and an immersive activity on Optima's VR campus reinforces each topic. Honors scholars go further — proving by contradiction, deriving the Laws of Sines and Cosines, and constructing tangents and inscribed polygons. The course assumes that learning to prove a claim, not merely compute an answer, is a skill worth practicing for its own sake and for every course in mathematics, science, and engineering that follows.

WHAT SCHOLARS WILL LEARN TO DO

- Prove and apply theorems about lines, angles, and triangles.
- Establish triangle congruence and similarity (SSS, SAS, ASA, AAS, AA, HL) both from Euclidean criteria and from rigid transformations.
- Reason logically: build truth tables, conditional statements, counterexamples, and proofs — including proof by contradiction.
- Analyze quadrilaterals and circles and justify their properties.

- Solve right triangles using trigonometric ratios, the Pythagorean Theorem, and the Laws of Sines and Cosines.
- Use coordinate geometry to classify figures and solve problems involving distance, midpoint, perimeter, and area.
- Apply transformations, and find area, surface area, and volume of two- and three-dimensional figures.
- Construct figures with a compass and straightedge, and write equations of circles.

Major Assessments & Milestones

Each module builds toward a summative assessment. Sample target dates below follow a full-year pacing; On-Demand scholars pace to their own enrolled term and Live sections follow the class schedule, keeping each milestone roughly in order.

Milestone	Course place	Sample target	What you complete
Semester 1 summatives	Modules 1–6 · Q1–Q2	by Dec 18 (Q2)	Module summative assessments for foundations, logical reasoning, congruency, similarity, trigonometry, and quadrilaterals.

Milestone	Course place	Sample target	What you complete
Constructions & proofs	Throughout · esp. M1, M2, M7	ongoing	Compass-and-straightedge constructions and written proofs submitted in Canvas, including honors constructions of tangents and inscribed polygons.
Semester 2 summatives	Modules 7–11 · Q3–Q4	by May 28 (Q4)	Module summative assessments for circles, measurement, coordinate geometry, transformations, and advanced applications.

Important Dates — 2026–2027

Aug 12, 2026	First day of school	**Sep 7**	Holiday (Labor Day)
Oct 9	Quarter 1 ends	**Oct 12**	Holiday (Columbus Day)
Nov 11	Holiday (Veterans' Day)	**Nov 23–27**	Thanksgiving break
Dec 18	Quarter 2 ends · fall semester ends	**Dec 21–Jan 1**	Winter break
Jan 5, 2027	Classes resume	**Jan 18**	Holiday (MLK Day)
Feb 15	Holiday (Presidents' Day)	**Mar 5**	Quarter 3 ends
Mar 26–Apr 5	Spring break	**Apr 6**	Classes resume
May 28, 2027	Quarter 4 ends · last day of school	**May 31**	Holiday (Memorial Day)

Assessment & Grading

45%
Independent practice & mid-module checks

20%
Constructions & proofs

25%
Module summative assessments

10%
VR campus activities

Most assignments allow at least two attempts, so scholars can review what they missed and try again to show learning gains. Summative Reviews are scored for completion; Summative Assessments are scored for correctness.

How This Course Is Delivered

This course is offered in two formats that share the same content, assignments, and expectations: an **On-Demand (asynchronous)** section you pace yourself, and a **Live (synchronous)** section that meets on a schedule on Optima's VR campus. The two overviews below explain how each one works. Find the one that matches your enrollment.

On-Demand (asynchronous) learning

In the On-Demand section you move through the course at your own pace within each quarter, guided by due dates that keep you on track. All assignments for a quarter open at its start and close at its end. Because the course is On-Demand, there is no grade penalty for missing an individual due date — but any assignment not completed by the end of the quarter receives a zero, and assignments submitted in the final 14 days of a quarter receive a 10% deduction, so steady pacing matters.

Attendance is measured by assignment completion. Per school policy, a scholar who completes no assignment for ten consecutive school days, without notifying the school of an absence in advance, may be dropped from the course. Completion means a sincere attempt — real effort on real work. Rushed work, wrong or corrupted files, irrelevant or nonsense submissions, AI-generated filler, and word-salad responses do not count.

👤 **You are never on your own.** A live facilitator grades your work, gives feedback, and hosts a weekly study hall and VR workshops.

Live (synchronous) learning

In the Live section you join scheduled class meetings on Optima's VR campus, where your teacher leads instruction in real time and you learn alongside classmates. The lessons, assignments, and assessments are the same as the On-Demand course; what differs is the pacing — a Live section moves through the material together on the class schedule rather than at each scholar's own pace, and class time is also used for discussion, questions, and guided practice with proofs and constructions.

Attendance is measured by presence and participation. You are expected to attend the scheduled live sessions, on time and ready to take part. If you must miss a class, notify the school in advance; your teacher will let you know how to make up the work and any material you missed. As in every Optima course, completing your work means a sincere attempt — real effort on real work — and assignments still open and close with each quarter.

🎧 **You meet regularly with a teacher.** The class gathers throughout the week in VR and on Microsoft Teams for live instruction and discussion.

Appropriate Use of AI

Optima scholars are preparing for a world shaped by artificial intelligence, and learning to use these tools with good judgment is part of that preparation. You are welcome to use AI tools to support your learning — to explain a concept a different way, check your understanding, or see another worked example — unless a particular assignment says otherwise. What AI may never do is the thinking or the work an assignment asks of you: a proof, a construction, a self-assessment, or a solution you submit must be

genuinely your own. Submitting AI-generated work as if it were your own is dishonest and, as noted above, counts as an insincere attempt. When you do use an AI tool on an assignment, say so briefly. Above all, build the underlying skill yourself — the goal is for you to be able to judge whether an AI's answer is any good, which you can only do if you can do the mathematics without it.

Learning in Virtual Reality

Optima is a virtual-reality school. Alongside your Canvas coursework, you will join immersive sessions on Optima's VR campus, where you meet teachers and fellow scholars, take part in live activities, and explore geometric ideas — building, measuring, and transforming figures in space — in ways you could not in an ordinary classroom. Each lesson includes a VR resource that reinforces its content. Keep your headset charged and ready for any scheduled VR sessions, and treat the VR campus with the same courtesy and seriousness you would a physical one. If you ever cannot get into VR for a session, tell your facilitator — there is an alternative way to take part so you do not fall behind.

Being an Optima Scholar

Optima's mission is an excellent, virtue-centered education, and virtue is at the center of this course as much as any other. Every scholar is expected to practice the seven virtues that shape life at Optima — **courage, courtesy, honesty, perseverance, self-government, service, and responsibility**. In practice that means: do your own honest work; treat everyone you meet in, through, or because of this course with respect, in writing, in discussion, and in VR; engage actively rather than passively; come to office hours to get help, meet other scholars, and get to know your facilitator; and take responsibility for your own pacing and progress. Geometry rewards perseverance above all — a hard proof is meant to be wrestled with. You belong to a house and a community of scholars, not only to a course.

What You Need & Staying Connected

- A computer with reliable internet and access to **Canvas**.
- A **VR headset**, charged and ready for the immersive activities on Optima's VR campus.
- A **compass, straightedge, protractor, and pencils** for constructions — plus access to GeoGebra (linked in Start Here) for web-based constructions.
- **Microsoft 365** (Word, Excel) with OneDrive, for documents and work you create and submit.
- A notebook or binder for notes, sketches, and proof practice.

Staying connected. Updates about this course are posted on the course's Canvas page; school-wide updates are posted in your Homeroom Canvas course. Check both regularly. Your facilitator holds office hours and is glad to help with the material, to meet you, or to talk through your plans — come by, or reach out any time by email.

Geometry Honors · Florida course 1206320 · Full-year high-school course · Optima Academy Online, 2026–2027. Built from the course content (11 modules) and the board-approved 2026–2027 academic calendar. Milestone target dates reflect a sample full-year pacing; On-Demand scholars pace to their own enrolled term. School calendar dates may be adjusted by the school.